

支持向量机 SVM.

一、线性可分支持向量机.

1.1 寻找一个超平面, 使两类样本之间间隔 (Margin) 最大.

1.2. $D = \{x_i, y_i\}_{i=1}^n, x_i \in \mathbb{R}^d, y_i \in \{-1, +1\}$.

超平面: $f(x) = w^T x + b = 0$

分类决策: $h(x) = \text{sign}(w^T x + b)$.

1.3. $f_i = y_i(w^T x_i + b)$. 函数间隔.

$$\gamma_i = \frac{y_i(w^T x_i + b)}{\|w\|} = \frac{f_i}{\|w\|}.$$

证明: 设 x^* 为 x_i 的投影.

则 $d = \|x_i - x^*\|$ 且 $(x_i - x^*) \perp w$

故 $\exists k$ s.t. $x_i - x^* = kw$.

$\because x^*$ 在平面上, $w^T x_0 - w^T x^* = kw^T w$.

$$w^T x_0 + b = k\|w\|^2.$$

$$k = \frac{w^T x_0 + b}{\|w\|^2}$$

$$d = \|x_0 - x^*\| = |k| \cdot \|w\| = \left| \frac{w^T x_0 + b}{\|w\|_2} \right| = \frac{|w^T x_0 + b|}{\|w\|_2}$$

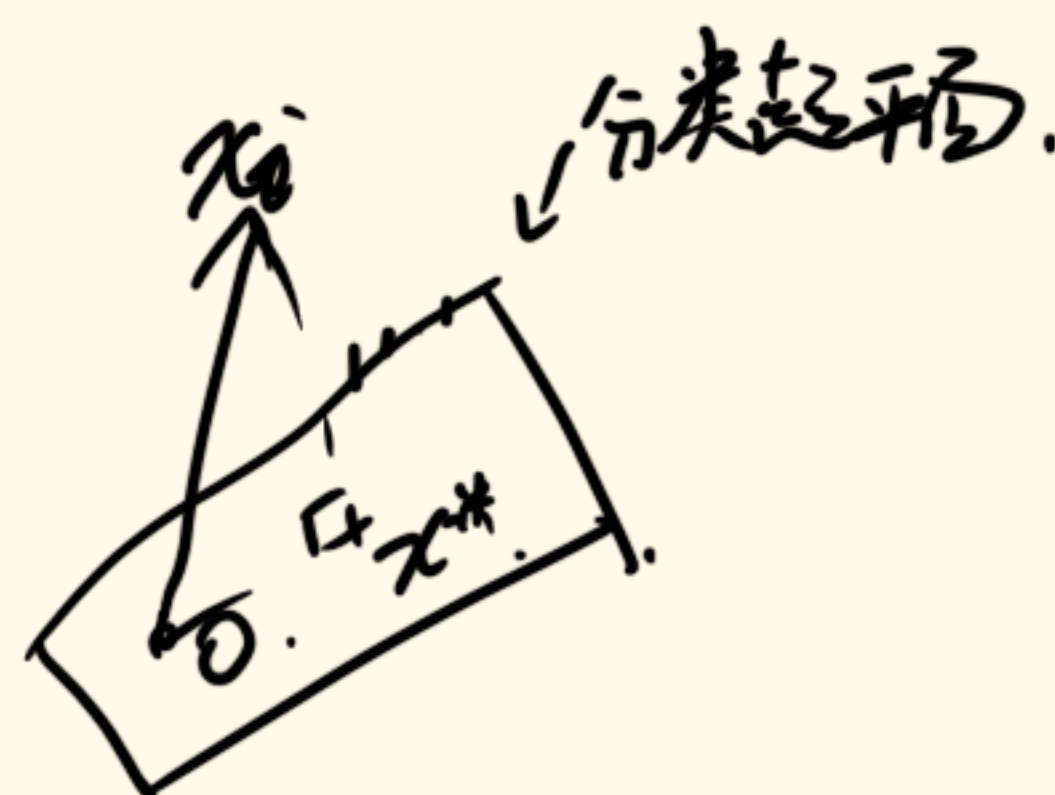
1.4. 故我们得到优化问题.

$$\max_{w, b} \gamma$$

s.t. $y_i(w^T x_i + b) \geq \gamma, i = 1, \dots, n$

由于 $\gamma = \frac{f}{\|w\|}$, 且间隔可缩放 (w, b 同比缩放, 超平面不变)

我们因取 $\gamma = 1$



等价于: $\max_{w, b} \frac{1}{\|w\|}$

s.t. $y_i(w^T x_i + b) \geq 1, i=1, \dots, n.$

进一步等价: $\min_{w, b} \frac{1}{2} \|w\|^2$

s.t. $y_i(w^T x_i + b) \geq 1, i=1, \dots, n.$

二、凸优化问题与KKT条件

2.1 凸优化识别:

凸集: C 为凸 $\Leftrightarrow \forall x, y \in C, \theta \in [0, 1],$ 有 $\theta x + (1-\theta)y \in C.$

凸函数: f 为凸 $\Leftrightarrow \forall x, y \in \text{dom}(f), \theta \in [0, 1],$ 有
 $f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y).$

同样地, 函数上两点连线,
该线在曲线上方.

SVM优化是凸:

① $\frac{1}{2} \|w\|^2$ 为凸函数.

② $y_i(w^T x_i + b) - 1 \geq 0$ 是仿射函数, 是凸约束.

2.2 拉格朗日函数

原始问题: $\min_{w, b} \frac{1}{2} \|w\|^2.$

s.t. $1 - y_i(w^T x_i + b) \leq 0, i=1, \dots, n.$

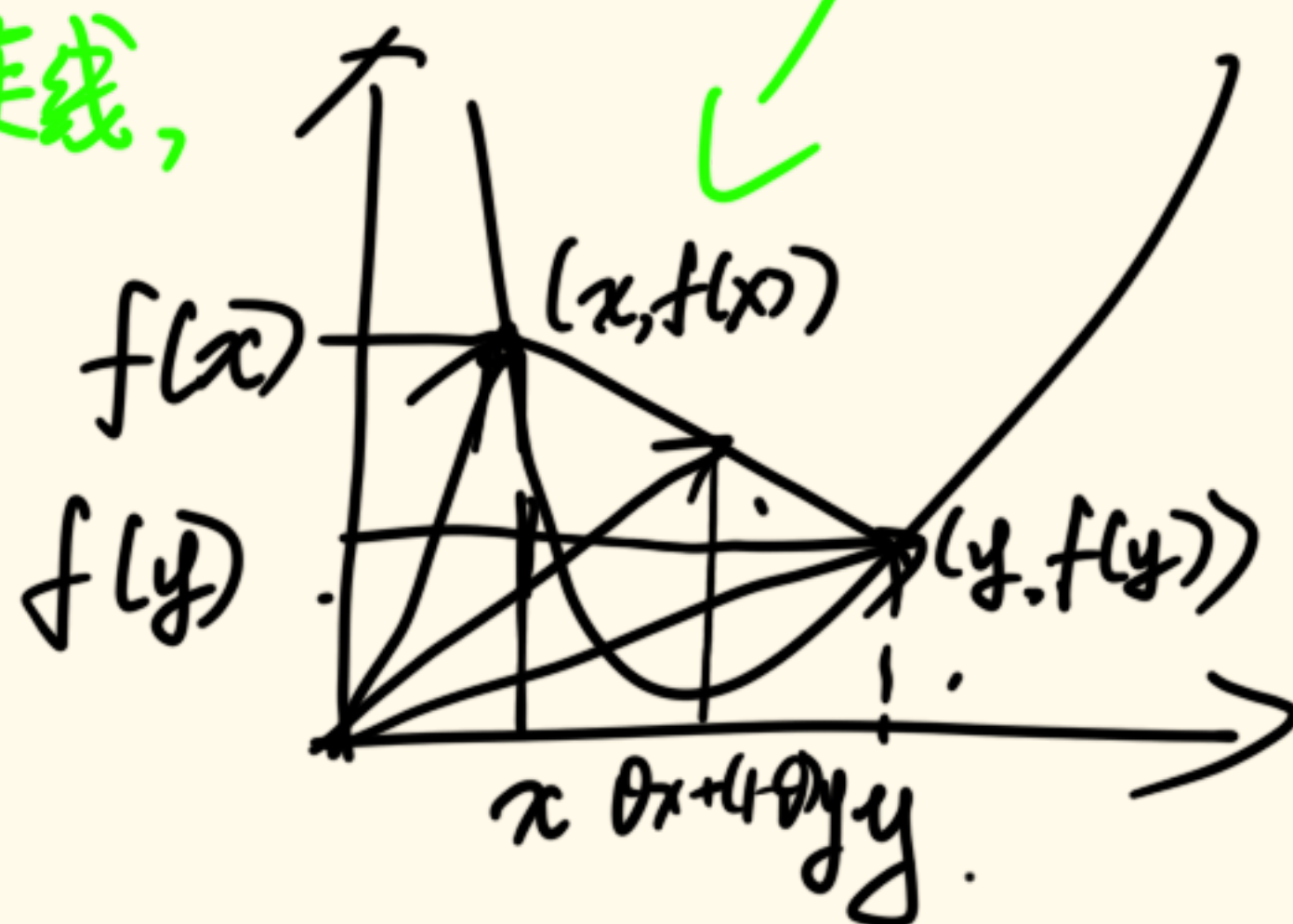
拉格朗日函数: (引入拉格朗日乘子 $\alpha_i \geq 0$).

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 + \sum_{i=1}^n \alpha_i [1 - y_i(w^T x_i + b)].$$

$$\alpha = [\alpha_1, \dots, \alpha_n]^T \geq 0$$



别忘了 弦比切线公式.



$$\begin{cases} \frac{\partial L}{\partial w} = w - \sum_{i=1}^n \alpha_i y_i x_i = 0 \\ \frac{\partial L}{\partial b} = -\sum_{i=1}^n \alpha_i y_i = 0 \end{cases} \Rightarrow \begin{cases} \sum_{i=1}^n \alpha_i y_i x_i = w \\ \sum_{i=1}^n \alpha_i y_i = 0 \end{cases}$$

将 $w = \sum_{i=1}^n \alpha_i y_i x_i$ 代入 L .

$$\begin{aligned} L &= \frac{1}{2} \|w\|^2 + \sum_{i=1}^n \alpha_i [1 - y_i (w^T x_i + b)] \\ &= \frac{1}{2} \left\| \sum_{i=1}^n \alpha_i y_i x_i \right\|^2 + \sum_{i=1}^n \alpha_i [1 - y_i \left(\left(\sum_{j=1}^n \alpha_j y_j x_j \right)^T x_i + b \right)] \\ &= \frac{1}{2} \left(\sum_{i=1}^n \alpha_i y_i x_i \right)^T \left(\sum_{j=1}^n \alpha_j y_j x_j \right) + \sum_{i=1}^n \alpha_i - \sum_{i=1}^n \alpha_i y_i x_i^T \sum_{j=1}^n \alpha_j y_j x_j - \sum_{i=1}^n \alpha_i y_i b \\ &= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j + \sum_{i=1}^n \alpha_i - \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j - b \sum_{i=1}^n \alpha_i y_i \\ &= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j \end{aligned}$$

KKT条件:

$$\begin{cases} ① \begin{cases} w = \sum_{i=1}^n \alpha_i y_i x_i \\ \sum_{i=1}^n \alpha_i y_i = 0 \end{cases} \\ ② 1 - y_i (w^T x_i + b) \leq 0, i=1, \dots, n \\ ③ \alpha_i \geq 0, i=1, \dots, n \\ ④ \alpha_i [1 - y_i (w^T x_i + b)] = 0 \end{cases}$$

稳定性.

梯度点 > "最优解" 存在.

原始可行性

满足所有约束

对偶可行性

必须非负, 因为不等式约束

对偶度量可行

"抬高违规函数"

互补松弛性

一部分为0, 一部分不为0.

关于互补松弛性: 若 $\alpha_i = 0$, 则 i 为支持向量.

若 $\alpha_i > 0$, 则 $1 - y_i (w^T x_i + b) = 0$, i 为支持向量.

三、线性不可分情况:

3.1. 引入松弛变量.

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

C. 惩罚参数, 控制间隔宽度与误差

$$\text{s.t.} \begin{cases} y_i(w^T x_i + b) \geq 1 - \xi_i, \quad i=1, \dots, n. \\ \xi_i \geq 0, \quad i=1, \dots, n \end{cases}$$

引入 α 与 μ 的约束

$$L(w, b, \xi, \alpha, \mu) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i + \sum_{i=1}^n \alpha_i [1 - \xi_i - y_i(w^T x_i + b)] - \sum_{i=1}^n \mu_i \xi_i$$

$$\begin{cases} \frac{\partial L}{\partial w} = 0 \Rightarrow w = \sum_{i=1}^n \alpha_i y_i x_i \\ \frac{\partial L}{\partial b} = 0 \Rightarrow \sum_{i=1}^n \alpha_i y_i = 0 \\ \frac{\partial L}{\partial \xi_i} = 0 \Rightarrow C - \alpha_i - \mu_i = 0 \Rightarrow \alpha_i = C - \mu_i \end{cases}$$

由于 $\mu_i \geq 0$,
 $\alpha_i \leq C$

对偶问题:

$$\max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j$$

$$\text{s.t.} \begin{cases} 0 \leq \alpha_i \leq C, \quad i=1, \dots, n \\ \sum_{i=1}^n \alpha_i y_i = 0 \end{cases}$$

$\alpha_i = 0$ 样本正确分类, 不在间隔边界上
 $0 < \alpha_i < C$: 样本在边界上, $\mu_i \neq 0$, $\xi_i = 0$.
则 $y_i(w^T x_i + b) = 1$.
 $\alpha_i = C$: 在间隔内或误分类.

KKT 条件.

$$\begin{cases} \frac{\partial L}{\partial w} = w - \sum_{i=1}^n \alpha_i y_i x_i = 0 \\ \frac{\partial L}{\partial b} = -\sum_{i=1}^n \alpha_i y_i = 0 \\ \frac{\partial L}{\partial \xi_i} = C - \alpha_i - \mu_i = 0 \end{cases}$$

① 稳定性条件.

$\alpha_i + \mu_i = C$ 用于区分点的位置.

$$\begin{cases} y_i(w^T x_i + b) \geq 1 - \xi_i \\ \xi_i \geq 0 \end{cases} \quad \textcircled{2} \text{原始可行性}$$

$$\begin{cases} \alpha_i \geq 0 \\ \mu_i \geq 0 \\ \alpha_i + \mu_i = C \end{cases} \Rightarrow 0 \leq \alpha_i \leq C \quad \textcircled{3} \text{对偶可行性}$$

$$\begin{cases} \alpha_i [y_i(w^T x_i + b) - 1 + \xi_i] = 0 \\ \mu_i \xi_i = 0 \end{cases} \quad \begin{array}{l} \text{间隔约束 (是不是支持向量)} \\ \text{互补松弛性} \end{array}$$

松弛度量 (要不要惩罚)

四. 核方法与非线性SVM.

$$\varphi: \mathbb{R}^d \rightarrow \mathbb{R}^m \quad f(x) = w^T \varphi(x) + b$$

$$w = \sum_{i=1}^n \alpha_i y_i \varphi(x_i).$$

$$\text{故 } f(x) = \sum_{i=1}^n \alpha_i y_i \varphi(x_i)^T \varphi(x) + b.$$

核技巧: $\mathbb{R}^m$ 定义 $K(x_i, x_j) = \varphi(x_i)^T \varphi(x_j)$.

则对偶问题变为: $\max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j K(x_i, x_j)$

优势: 无需显式计算 $\varphi(x)$, 只需计算核函数.

常用: 线性核: $K(x_i, x_j) = x_i^T x_j$.

多项式核: $K(x_i, x_j) = (\gamma x_i^T x_j + r)^d$.

高斯核: $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$

Sigmoid核: $K(x_i, x_j) = \tanh(\gamma x_i^T x_j + r)$.

有效性 (Mercer条件) $K = [K(x_i, x_j)]_{n \times n}$ 必须对称半正定.

例: $x=(x_1, x_2)$ $z=(z_1, z_2)$.

$\varphi(x)=(x_1^2, x_2^2, \sqrt{2}x_1x_2, \sqrt{2}x_1, \sqrt{2}x_2, 1)$ 维, 很困难.

$$\langle \varphi(x), \varphi(z) \rangle = (x_1z_1 + x_2z_2 + 1)^2$$

则核函数为 $k(x, z) = (x^T z + 1)^2$.

对于高斯核 RBF, 令 $\gamma = \frac{1}{2\sigma^2}$.

$$k(x, z) = e^{-\gamma \|x - z\|^2}.$$

由 Taylor 展开, $e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!}$.

$$\|x - z\|^2 = \|x\|^2 - 2x^T z + \|z\|^2.$$

$$k(x, z) = e^{-\gamma(\|x\|^2 - 2x^T z + \|z\|^2)} = e^{-\gamma\|x\|^2} \cdot e^{2\gamma x^T z} \cdot e^{-\gamma\|z\|^2}.$$

$$\text{令 } u = 2\gamma x^T z.$$

$$e^u = \sum_{n=0}^{\infty} \frac{(2\gamma)^n}{n!} (x^T z)^n.$$

$$\therefore (x^T z)^n = (x_1z_1 + x_2z_2 + \dots + x_dz_d)^n.$$

$$d=2 \text{ 时, } n=1: x_1z_1 + x_2z_2.$$

$$n=2: x_1^2z_1^2 + 2x_1x_2z_1z_2 + x_2^2z_2^2.$$

$$n=3: 1 \quad 3 \quad 3 \quad 1$$

$$n=\infty: \text{无限多项}.$$

维度高样本少 $\rightarrow$ 线性核.

维度低样本多 $\rightarrow$ 高斯核.

有先验知道是多项式 $\rightarrow$ 多项式核.

什么都不知道 $\rightarrow$ 高斯核.

$$\begin{aligned} \text{合起来后, } k(x, z) &= e^{-\gamma\|x\|^2} e^{-\gamma\|z\|^2} \sum_{n=0}^{\infty} \frac{(2\gamma)^n}{n!} (x^T z)^n \\ &= \sum_{n=1}^{\infty} \varphi_n(x) \varphi_n(z) = \langle \varphi(x), \varphi(z) \rangle \end{aligned}$$

五. SVM 的求解:

SMD 算法: 先选两个样本 α_1, α_2 , 且 $\sum_{i=1}^n \alpha_i y_i = 0$ 等:

线性等式约束 $\alpha_1 y_1 + \alpha_2 y_2 = \zeta$, $\zeta = -\sum_{i=3}^n \alpha_i y_i$ 是常数

盒约束 $s.t. 0 \leq \alpha_i \leq C$
直到收敛/超时为止.

例: $x_1=0$ $y_1=1$ (正) $k = x_2$
 $x_2=2$ $y_2=1$ (正) $C=10$
 $x_3=1$ $y_3=-1$ (负) 初始: $\alpha_1 = \alpha_2 = \alpha_3$

第一轮: 固定 α_1 , $\alpha_2 - \alpha_3 = \alpha_1 y_1 = 0$ $\alpha_2 = \alpha_3$.

$$E_2 = f(x_2) - y_2 = 0 - 1 = -1 \quad \text{当前轮次下的模型.}$$

$$\bar{E}_3 = f(x_3) - y_3 = 0 - (-1) = 1$$

$$\eta = k_{22} + k_{33} - 2k_{23} = 4 + 1 - 4 = 1 \quad \text{核函数.}$$

$$x_3^{unc} = \alpha_3^{old} + \frac{y_3(E_2 - \bar{E}_2)}{\eta} = 0 + \frac{(-1)(-1 - 1)}{1} = 2$$

算 $[L, H]$: $y_2=1, y_3=-1$

$$L = \max(0, \alpha_3 - \alpha_2) = 0, H = \min(C, C + \alpha_3 - \alpha_2) = 10$$

$$\alpha_3^{new} = 2.$$

$$\text{由 } \alpha_2 = \alpha_3, \alpha_2^{new} = 2$$

第一轮结果: $\alpha = [0 \ 2 \ 2]$.

$$\text{约束: } 1 \cdot 0 + 1 \cdot x_2 + 1 \cdot 2 = 0. \quad \checkmark$$

$$\text{范围: } 0 \leq 2 \leq 10 \quad \checkmark$$

当为线性可分问题时, $C = +\infty$.

